# Present and Future, Challenges of High Bandwith Memory (HBM)

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***Abstract*— In recent years, the demand for fast and high-capacity memory has surged due to the emergence of generative AI models such as GPT. To address this need, High Bandwidth Memory (HBM) has rapidly risen as a critical memory solution. This paper aims to elucidate why HBM plays a pivotal role in the AI industry and discusses the imminent challenges that need to be overcome in the development of HBM memory. By exploring the significance of HBM in the context of AI applications and outlining the pressing challenges ahead, this paper contributes to understanding the importance of HBM in meeting the memory requirements of advancing AI technologies.**

***Index Terms*—HBM, Stacked DRAM, TSV, 2.5D SiP, High bandwidth DRAM, MR MUF, AI Memory**

## I. Introduction

With the advent of digital transformation and the emergence of ultra-large artificial intelligence (AI) models, there has been a significant increase in the generation of massive datasets, leading to a growing demand for rapid computation. The proliferation of new technologies such as AI, autonomous driving, and smart services has further fueled the demand for high-performance computing capabilities to process the vast amounts of data. In particular, with the intensification of competition in the realm of ultra-large AI models exemplified by ChatGPT, the necessity for ultra-high-performance computing has become increasingly apparent. Consequently, there has been a notable emergence of the need for memory solutions optimized for high-capacity data processing [1]. (Fig. 1[2])

The von Neumann architecture, consisting of separate processing units (CPU/GPU) and memory, faces challenges due to the limitation of latency in data movement, resulting in system performance degradation. High Bandwidth Memory (HBM) has gained significant attention as a potential solution to alleviate these data movement bottlenecks [3]-[7]. With its high bandwidth, HBM enables the simultaneous transfer of large volumes of data and rapid processing, making it a focal point of interest for industries requiring high-performance computing capabilities. Moreover, HBM holds substantial potential to advance high-performance computing (HPC) and has established itself as the memory solution of choice for AI/ML and other high-performance computing workloads, providing unparalleled memory bandwidth in compact, high-capacity installation spaces.

The journey of HBM began with speeds of 128GB/s and a maximum of 4 stacks of 2Gb devices, steadily increasing in speed through continuous development. This culminated in the recent production of HBM3E, boasting bandwidths exceeding 1.18TB/s within less than a decade. HBM has since become a cornerstone for entirely new types of HPC accelerators introduced to the market. (Fig2)

However, maintaining these increased speeds has become progressively challenging due to the fundamental limitations inherent in DRAM cells. Additionally, challenges such as heat generation resulting from speed and capacity increases, as well as the necessity to increase capacity within constrained structures, have arisen.

In this paper, we aim to examine the challenges faced by HBM, propose potential solutions, and envision the future evolution of HBM

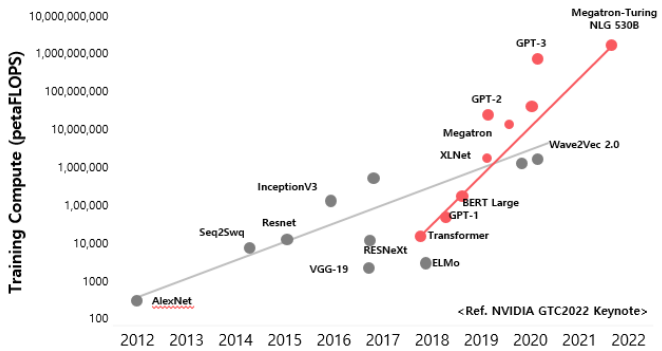


Fig. 1. Computational Requirements for Training Transformers.

TABLE 1. Key Feature Comparisons of HBM Generations

| Category | Item | HBM1 | HBM2 | HBM2E | HBM3 | HBM3E | HBM4 |
|---|---|---|---|---|---|---|---|
| | Time | 2014 | 2018 | 2020 | 2022 | 2024 | 2026 |
| General | Die Density | 2Gb | 8Gb | 16Gb | 16Gb | 24Gb | 24Gb |
| | Max. Bandwidth | 128GB/s | 0.3TB/s | 0.5TB/s | 0.7TB/s | 1.18TB/s | 1.65TB/s |
| Configuration | Max. Stack Height | 4HI | 8HI | 8HI | 12HI | 12HI | 16HI |
| | Capacity | 1GB | 8GB | 16GB | 24GB | 36GB | 48GB |
| | Total IO/Cube | 1024 | 1024 | 1024 | 1024 | 1024 | 2048 |
| Power | vDDC | 1.2V | 1.2V | 1.2V | 1.1V | 1.1V | 1.05V |
| | vPPE | 2.5V | 2.5V | 2.5V | 1.8V | 1.8V | 1.8V |
| | vDDQ | 1.2V | 1.2V | 1.2V | 1.1V | 1.1VV | 0.8V |


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## II. PRESNET HIGH-BANDWIDTH MEMORY (HBM)

The High Bandwidth Memory (HBM) has been developed since its introduction in 2014 to address the increasing demand for high-performance, high-capacity, low-latency memory required for High Performance Computing (HPC), High Performance Graphics Processing Units (GPUs), and Artificial Intelligence (AI). With each new generation, such as HBM2, HBM2e, HBM3, and the recently commenced mass production of HBM3E, bandwidth and capacity requirements have continued to rise. (TABLE 1)

HBM3E, developed by SK hynix, is currently the highest specification HBM available worldwide (Fig 2). It can process over 1.18 terabytes (TB) of data per second. Compared to the previous generation, HBM3, its speed has increased by 1.4 times, and its capacity by 1.5 times. While HBM3 offered packages with 16GB in 8-stack(8H) and 24GB in 12-stack(12H) configurations, HBM3E provides a capacity of 24GB in an 8-stack configuration. SK hynix is preparing for future packages with 36GB capacity in a 12-stack configuration. With the increase in HBM speed and capacity, the persistent issue of heat generation has been addressed by applying the Advanced MR-MUF technology, resulting in a 10% improvement in heat dissipation performance compared to the previous generation.

HBM3E also offers backward compatibility, allowing new products to be used with existing CPUs and GPUs without the need for design specification changes from the previous versions. Similarly, systems designed with HBM3 can seamlessly adopt new products without the need for design or structural changes.

However, due to the unchanging basic physics of DRAM cells, maintaining these increased memory transfer speeds has become increasingly challenging. As a result, for HBM4, major memory manufacturers are planning to support more substantial changes to HBM technology by adopting a much wider 2048-bit memory interface.

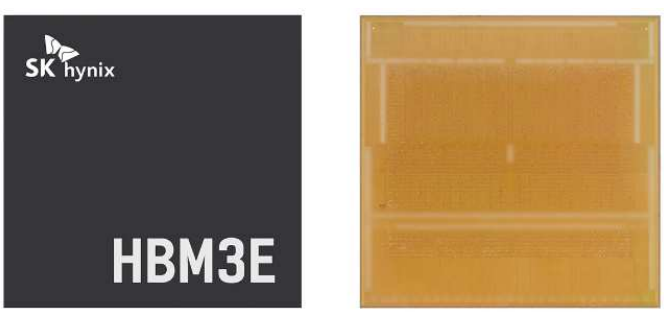


Fig. 2. HBM3E Image

## III. HBM CHALLENGES AND SOLUTION

### A. HBM Bandwidth expansion

The most crucial aspect of HBM, as evident from its name, is its High Bandwidth Memory. The bandwidth of HBM has approximately doubled every two years over the past decade (TABLE 1).

However, increasing IO speed has become increasingly difficult, and the channel characteristics of the silicon interposer have become problematic. To address this issue, wider IOs can be utilized, or operating frequencies can be increased to expand the bandwidth. The next generation of HBM will utilize 2048 IOs. Although the pin speed is slightly slower than the previous generation, the total bandwidth is significantly higher. This approach allows for further enhancement of IO speed and channel characteristics, ensuring scalability (Fig. 3).

Additionally, in previous iterations, base dies were manufactured using DRAM processes to reduce manufacturing costs. However, in the next generation of HBM, base dies will be integrated with advanced logic technology to improve performances.

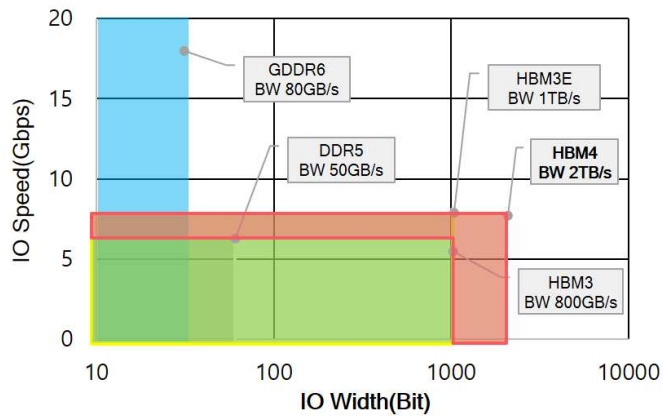


Fig. 3. IO speed and width of HBM and various DRAM applications

### B. Power and Thermal Challengease

Another challenge of HBM is that its 2.5D structure traps heat, leading to thermal degradation of the DRAM near the GPU. GPUs generate vast amounts of data (terabits of data per second) through interfaces and perform billions of operations per second, resulting in significant heat generation [8]. (fig.4)

Logic-based dies not only improve bandwidth but also reduce power consumption. Traditional HBM base dies consume approximately 40% of the total power. By utilizing logic processes, the supply voltage can be reduced to less than 0.8V from 1.1V, achieving a 50% power savings. A decrease in power consumption also addresses heat-related issues, as a 20% reduction in power consumption corresponds to a 20% reduction in heat generation.

Furthermore, SK hynix's proprietary advanced packaging solution known as MR-MUF supports device cooling. This technology allows for the integration of additional TSVs, resulting in further power savings. SK hynix has successfully developed and mass-produced HBM2E and HBM3 products using MR-MUF and has developed advanced MR-MUF methods for implementing the structure of 12H, a high-stack product, in the future. (Fig.5)

MR-MUF is a material with excellent heat dissipation properties. In future products, further improvements in thermal

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characteristics are expected by applying materials like hybrid bonding.

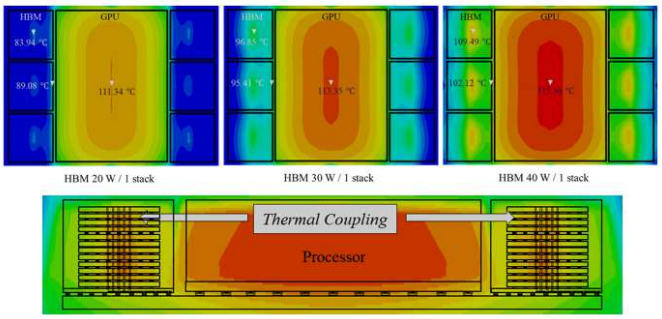


Fig. 4. Thermal Analysis of HBM-GPU Module [9]

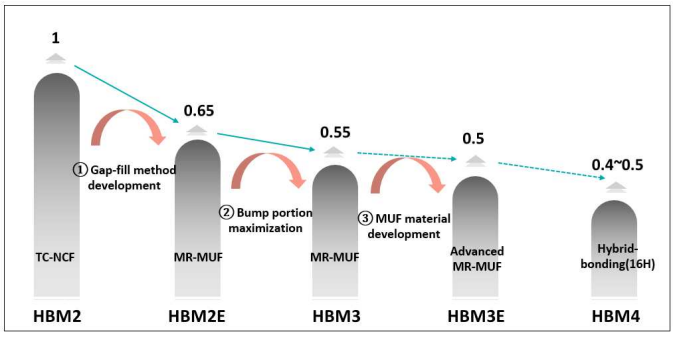


Fig. 5. Thermal resistance according to MR-MUF process improvement

However, power integrity remains another challenge. Between the top core die of HBM and the SoC, numerous components generate significant power noise as power is supplied through complex networks, including PCBs, substrates, silicon interposers, base dies, and stacked core dies. Moreover, to reduce power noise, additional power TSVs are essential. Past attempts to add more power TSVs to HBM products were challenging because TSVs had to avoid DRAM cell areas. However, through proprietary design technology, we have succeeded in increasing the number of power TSVs by five times compared to previous efforts. The impact varies for each circuit, but the overall range of improvement in the power delivery network (PDN) is from 16% to 75%. (Fig.6) [10]

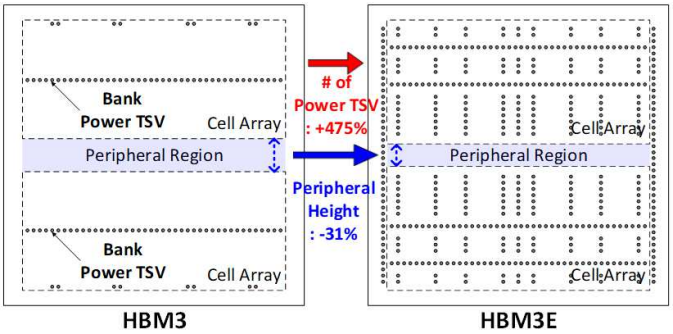


Fig. 6. Power TSVs spread scheme for improving PDN @HBM3E

*C. Increasing DRAM Density*

There are two methods to achieve this: integrating more DRAM cells into a single core die or stacking more core dies in a stack. The die density of the initial generation HBM2 was 8Gb, but we have increased it to 24Gb, reaching 36GB for 12H stack and 48GB for 16H stack. However, due to the challenges of technology scaling, die density increase became slower, and thin die handling is required for stacking. (Fig. 7)

The height of the HBM stack is limited to 750um, so additional issues arise as more stacks require thinner core dies. The thickness of the core die is approaching 30um, which is one-third the thickness of a human hair (approximately 100um). Additionally, placing more HBM on the silicon interposer is limited by the size of the silicon interposer, processing tools, and production yield.

Fig. 8 shows the interposer size trend. While success has been achieved in increasing size, increasing size leads to lower yields and increased costs. Therefore, the industry is transitioning to alternative technologies [11].

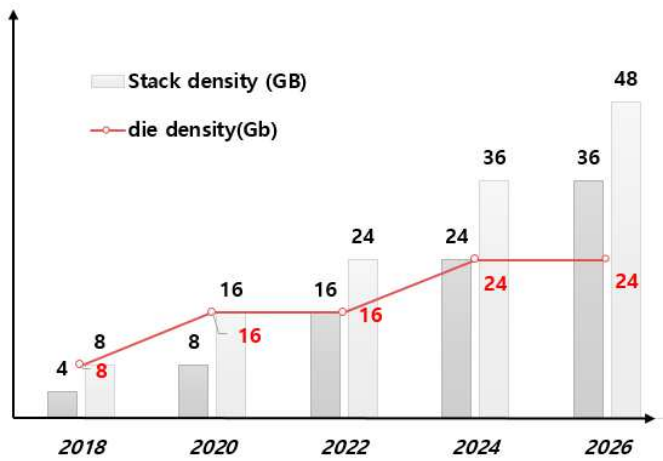


Fig. 7. HBM Density

LSI (Local Silicon Interconnect) has small silicon bridges that connect different ICs. These silicon bridges enable the size of SiP (System in Package) to be increased regardless of the limitations of silicon processing equipment.

Regarding stacking, the industry is also considering hybrid bonding technology, a technique for stacking multiple silicon dies without bumps. By reducing the bumps, it is possible to eliminate the gap between core dies without increasing the package height, allowing for more stacks with the same height.

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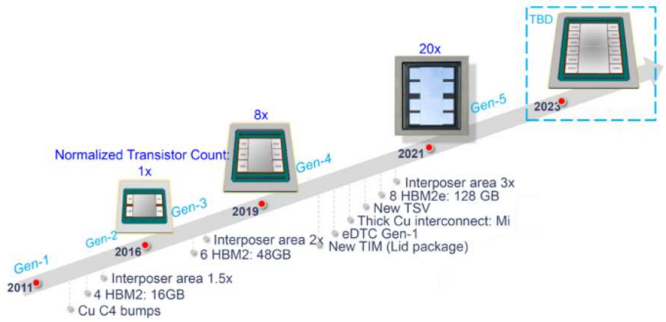


Fig. 8. TSMC's plan for 4x reticle size interposer [11]

## IV. Futuer High-Bandwidth Memory (HBM)

From a functional standpoint, traditional systems strictly separated computing and memory. However, with advancements in technology, computation can now occur in vertically stacked dies known as "near-memory processing." While this represents a noticeable improvement, there is still room for further efficiency enhancements. Implementing computing functionality in the core die results in a more efficient process known as "in-memory processing." Taking it a step further, calculations can be executed in an array of cells labeled "in-memory computation." Despite the potential efficiency gains, we have adhered to separate computing architectures due to the slower and less efficient nature of DRAM processes compared to logic processes. However, with logic processes integrated into the base die, there is no reason not to transition to near-memory processing. Therefore, we anticipate HBM products with advanced functionality integrated into the base die, opening up ways to integrate not only DRAM related circuits but also third-party IP or customer IP. Collaboration with customers on base die design is possible through shared process nodes. Consider leveraging proven high-speed SERDES IP, which can improve bandwidth without increasing IO or PHY size, and HBM can also be extracted from the package to the PCB, eliminating the size constraints of silicon interposers. Essentially, the bandwidth and density issues mentioned earlier can be addressed through high-speed SERDES IP solutions.

## V. Conclusion

In conclusion, the increasing demand for high-performance computing driven by digital transformation and the rise of ultra-large AI models has highlighted the importance of memory semiconductor solutions optimized for high-capacity data processing. High Bandwidth Memory (HBM) has emerged as a critical solution to address data movement bottlenecks, offering high bandwidth and low latency. Despite challenges such as maintaining increased speeds and addressing heat generation, HBM has continuously evolved, with recent advancements like HBM3E providing significant improvements in speed, capacity, and heat dissipation. To overcome technical challenges, future developments in HBM technology focus on wider IOs, integration with advanced logic technology, and advancements in thermal management. Additionally, the transition towards near-memory processing and collaboration with customers on base die design are expected to further enhance the efficiency and scalability of HBM solutions. Leveraging high-speed SERDES IP solutions can also address bandwidth and density issues, ensuring continued advancements in high-performance memory semiconductor solutions.

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